

SEDAR 35: Caribbean Red Hind

Data Workshop

No specific recommendations were provided.

Assessment Process

In the short term, US Caribbean stock assessments will continue to rely on mean-length estimation given the data currently available. The ability to use the mean-length estimator is contingent upon having length-frequency data that are temporally consistent and representative of the population and upon having reliable estimates of life history parameters, in particular, the von Bertalanffy parameters. Efforts should be made to review the current TIP sampling structure in Puerto Rico and in the USVI to ensure sampling is representative. Studies on basic life history (e.g., age-growth relationships, length/age-at maturity) in the US Caribbean will greatly enhance the utility of the existing length-frequency data and should provide the greatest benefit to providing management advice in the short term. This should be placed as a top priority for key species.

Fishery-independent surveys should be considered as a top research priority for additional data collection. Fishery-independent surveys designed using a rigorous statistical framework will allow for the collection of species-specific catch and effort data that can be used to develop indices of abundance. Indices of abundance are used in stock assessments to inform models about how a population may be changing over time. Fishery-independent surveys can also be used to supplement existing programs by collecting age, length, weight, and reproductive data.

During the SEDAR 35 assessment workshop, and in previous assessment workshops in the US Caribbean, the fishermen from the USVI indicated that the size of landed fish is market driven for plate size fish. This may help to explain the relatively narrow size range of landed Red Hind. It also suggests that selectivity is dome-shaped, which violates the assumption of knife-edge selectivity in the mean-length model. One avenue of future research would be to expand the mean-length estimator to accommodate other selectivity patterns. Another avenue of research would be to quantify the selectivity patterns for the different gear types. During the data and assessment workshops, the Panel could not quantify discard rates nor could they ascertain the level of discard mortality. If discard mortality of larger fish is significant, the violation of the selectivity assumption may be moot. Efforts should be made to quantify discard and discard mortality rates for the US Caribbean fisheries.

Lastly, under the current management regime all US fisheries must be managed by annual catch limits (ACLs). In an ideal scenario, ACLs would be developed from estimates of abundance and sustainable yield. The mean length estimator does not provide these metrics. As such, it is essential that continued efforts to improve the data collection of fishery-dependent catch and effort statistics be made so that traditional biomass-based assessment approaches can be employed. Continued efforts to collect

species-specific catch statistics will also be important in moving towards more traditional assessment approaches and for more precise monitoring of ACLs.

Review Workshop

Research Recommendations from the CIE Reviewer Reports (Term of Reference 6)

Cardinale Recommendations

The Assessment team provides an exhaustive list for future data to be collected, which would notably improve the capability of assessing the status of the Caribbean red hind stock. However, I consider that the description of the additional research and future monitoring is not exhaustively presented and it could have been much more detailed and comprehensive.

The reviewer agrees with the Assessment team that priority should be to given to derive data which allows movement towards more traditional assessment approaches. However, the reviewer also considers that this could in part already be pursued by the Assessment team using other methods than the mean length analysis (see ToR 2 and ToR 3).

Additional Recommendations

- A virtual population analysis (VPA), assuming a steady state and combining different gears, should be used for selected combination of years, areas and gears in a future assessment of Caribbean red hind.

Dowling Recommendations

The research recommendations provided by the Assessment Workshop are:

-(top priority) Undertake studies on basic life history (e.g. age--growth relationships, length/age at maturity).

- Agree, in so much as these should reduce existing uncertainty – but are these realistic given the existing capacity? Why are not previous studies considered representative? Are there existing studies for the same species elsewhere that may be helpful?

-Review the current TIP sampling structure to ensure sampling is representative.

- Agree -but “representative” in what sense? Temporally, spatially, of the size structure of the total fished population, of the total fishing effort?
- I think this should rate as a higher priority than undertaking fishery-independent surveys. The priorities should be immediately focused on improving the input to, and outcomes, the existing assessment approach.

-top priority) undertake fishery-independent surveys that enable the development of abundance indices, and that collect age, length, weight and reproductive data.

- Fair enough – but again, are these realistic given the existing capacity?
- Moreover, this recommendation should be made in the context of the evaluations of the existing fishery independent data and/or survey protocols (e.g. the Mona Island and Abrir la Sierra (DW03) protocols and data had potential had the time series been longer).

-To expand the mean-length estimator to accommodate other selectivity patterns.

- I think this is an excellent recommendation.

-To quantify the selectivity patterns for the different gear types.

- I agree that this needs to be resolved, especially given the assumption of knife-edged selectivity underpinning the per-recruit analyses.

-To attempt to quantify discard and discard mortality rates.

- Agree that this would be useful, but how could this be achieved? Quantifying discarding is notoriously difficult.

-To continue to improve the data collection of fishery-dependent catch and effort statistics so that traditional biomass-based assessment approaches can be employed (and hence annual catch limits determined and monitored).

- I agree that this is a key priority.
- However, there is presumably no way to improve the quality of the historical catch and effort statistics, so the issue is also one of how best to work with the existing data.
- While it may be ideal to develop ACLs from estimates of abundance and sustainable yield, these are often unavailable. This does not preclude ACLs from being set. ACLs may be determined using simple empirical approaches, while acknowledging the increased risk associated with less information and certainty.

Clearly denote research and monitoring that could improve the reliability of, and information provided by, future assessments.

There are two arenas for research and monitoring. One is around improving the reliability and usefulness of the current assessment approach. This equates to narrowing the range of mortality estimates and so increasing the certainty around the probability of overfishing (and to introducing an overfished reference point threshold). The other is around improving the

quality of information available into the future such that alternative approaches may be permitted.

Against the current assessment approach, I agree that

- Reviewing the TIP length data for representativeness (temporally, spatially, of the size structure of the total fished population, of the total fishing effort) is important. This should include a careful consideration of the TIP length data in the context of the other available length frequency information.
- Whether by improving the understanding of life history, and/or by reviewing the available information and parameters and weighting or narrowing these to a more plausible subset, working to reduce the range of mortality estimates is also important.

In addition, monitoring and research to resolve uncertainties around:

- -Stock structure in the context of the three regions considered (whether by genetic testing (expensive), tagging studies (expensive), or considering spawning migration (per RD09) and larval transport data (per RD06), and/or studies of dispersal, mixing and stock structure from similar species elsewhere) (i.e. are we treating the three regions as three separate stocks, to which different stock statuses and hence difference management apply?) and
- -Gear/fleet reconciliation within regions (i.e. what is the extent of overlap of the fishable sizes targeted/captured by the different gear/fleet types? To what extent can the data from each be combined?)

should be prioritized, so that the probabilities of overfishing are useful in a management context. Currently there are six sets of probabilities (3 regions x 2 overfishing threshold reference points), with each gear/fleet considered to contribute equally to the frequency distributions used to determine the probabilities. Ideally, there should be a recommended preferred reference point (that which is more precautionary, in the absence of other information), a better quantitative articulation of the extent to which the information from each gear/fleet contribute to the overall frequencies on which the probabilities of overfishing are based, and an increased confidence of how these should be applied in the context of what is understood about stock structure.

Additional work to determine:

- A suitable reference point corresponding to an overfished stock status
- A target reference point that could underpin management decision/control rules is also recommended.

Against improving the quality of information available into the future, such that alternative approaches may be permitted

- There is no clear indication given as to whether formal logbook reporting is possible (or exists), but it seems that the best means of obtaining uniform catch and effort data on which alternative assessments may be based (e.g. simple production models).
- A reconciliation of the fishery independent approaches should be undertaken with a view to recommending a monitoring program that will optimize the utility of the information obtained. (To what extent would the protocol described in RD01 be sufficient?) Recommendations should be pragmatic given the available resources and capacity.

Meanwhile, I encourage

- -Avoiding tossing out data for the sake of being overly Puritan. Even if time series of CPUE, for example, are not considered of adequate quality to enable a formal stock assessment, the data may be useful in informing simpler, more empirical assessments (e.g. Froese 2004; Dowling et al. 2008; Prince et al. 2012; Edwards et al. 2012; Erisman et al. 2014). At the very least, they give some notion of historical high catches, and size-based catch rates.
- -At least attempting to fit a production model to the two sets of standardized CPUE (AW01; DW04). Even if there proves to be inadequate contrast in the data, the attempt to use it in an assessment would still place emphasis on what is needed from future data collection protocols.

Finally, I strongly encourage that more effort be dedicated to considering approaches to developing ACLs. The Assessment Workshop avoided developing ACLs because the mean length estimator “does not provide these metrics” and “in an ideal scenario, ACLs would be developed from estimates of abundance and sustainable yield”. However, the yield-per-recruit/spawner-per-recruit analyses provide F_{MSY} target reference point proxies that could be used in determining an ACL via simulated projections. Second, fisheries arguably do not require “traditional biomass based assessment approaches” in order to set ACLs. Catch time series, triggers (as reference point proxies) or reference points, and empirical decision/harvest control rules can all be used to set an interim ACL (e.g. Dowling et al. 2008; Prince et al. 2012; Dowling et al. 2014).

Provide recommendations on possible ways to improve the SEDAR process.

On the basis of the documents provided, I have the following recommendations:

-Link the Data Workshop Report more closely to the Assessment Workshop Report, so that

- Data are consistently summarized between each report, preferably via a commonly presented summary table
- There is improved clarity on how and whether data are used in the assessment. There is minimal detail regarding data in the Assessment Workshop Report. It would have been useful had the Data Workshop Report indicated whether and how each type/set of available data was used in the assessment, both as a summary sentence at the time of its presentation, and in an overall data summary table.

-Prior to circulating for review, cross-check reports to ensure that report reference lists are complete and that key papers are included as background reading (or at least links provided). Perhaps allowing slightly more time for completion of reports may assist with this.

-As a required part of the Assessment Workshop Report, provide historical context and past precedence for assessments previously undertaken. This was not provided in current reports. If not previous precedent exists, this should be explicitly stated.

Maravelias Recommendations

The following are some general suggestions and recommendations to improve the current status of the fishery.

A. Improve the fishery information management system. The Puerto Rico's fishery has been monitored through the Fisheries Statistics Project (FSP) continuously since 1967. The project aimed to provide fisheries data for the resources in the waters of Puerto Rico and scientific information to support management plans. Despite this FSP initiative, the lack of reliable official fishery statistics is evident and constitutes a considerable handicap for the assessments. It is important to improve the official state authority design, implementation and integration of the system to collect and compile statistical data from the entire national fisheries. This data collection system should ideally cooperate with other authorities e.g. the port authorities, the local customs offices, correspondents in municipalities and communities, villages. The primary objective should be to collect fishery-dependent info: catch, effort, discards, fleet, economic (cost, profit), social (e.g. employment, education) statistics. Following standard and common sampling protocols for all isles, fleets, gears, seasons and strata. Similar data, especially catches, effort, discards, costs and profits, can be collected regularly using onboard sampling, i.e. following the fishers during their fishing trips. This will provide more realistic data that could then be compared with port sampling, intercepts, TIP, logbooks.

B. Basic research could be promoted to study Red Hind biological parameters. This research preferably

may include: age, growth, feeding, length/age-at-maturity, and fecundity to provide the fundamental knowledge that will support future assessments.

C. Fishery-independent surveys should be carefully designed and carried out in order to provide scientifically sound information and data to support stock assessment, fishery conservation and management. These ideally should cover the distribution of key species (including Red Hind) in all three studied regions, i.e. Puerto Rico, St Thomas/St John and St Croix. Such scientific surveys will provide abundance and biomass estimates but also additional size distribution, maturity, spawning season and areas, scales or otoliths for age and growth studies, stomach contents, fecundity information and they can target early-life stages and adult parts of the population. In addition a number of auxiliary data can be collected, e.g., oceanographic, seabed substrate, information on essential fish habitat of the species. These fishery-independent surveys will provide complete catch records in the area. Commercial vessels often discard many species and especially small fish (< MLS: minimum landing size), whereas research vessel surveys can provide information on the total species composition and size range available to the gear. The scientific information and data that will be collected will increase long-term economic and social benefits from the fisheries resources in the area. Once established, these surveys should be carried out routinely to support scientific monitoring of the living marine resources (e.g. annually or bi-annually).

D. Following the required provisions of the Magnuson-Stevens Fishery Conservation and Management Act, a number of management reference points for species undergoing overfishing were established by the 2010 Caribbean Annual Catch Limit Amendment 3. The Annual Catch Limit (ACL) is currently the main management tool and US fisheries should aim to specify ACLs and accountability measures, AMs, to prevent ACLs from being exceeded. Fishery-dependent catch, effort and discards statistics are urgently required to follow these provisions. As a first step, catch-based methods can be implemented that require only catch information. Biomass dynamic models can also be applied providing catch and effort data will become available. However, scientific advice to fishery managers needs to be expressed in probabilistic terms to convey uncertainty about the consequences of alternative harvesting policies. One avenue for future stock assessment could be to build informative prior probability distributions (priors) for r , K , q , M , F . Expert knowledge and the available fishery datasets may prove useful in building such priors. Then using a simple biomass dynamic model fitted to catch rate data, a risk assessment approach can be applied to evaluate the potential consequences of alternative ACLs. The benefit for the fishery from a probabilistic modeling method would be that uncertainties would have been considered but also estimates of biological risks of alternative ACL-policy options will be provided. This may serve as a basis for providing precautionary fishery management advice given the high degree of uncertainty.

E. Design and carry out gear selectivity studies aiming to disclose species' selectivity patterns and improve resource exploitation. This coupled with discard estimates from the fleet statistics and onboard scientific sampling will allow the assessment of discard mortality.

F. Improve the effectiveness of external partnerships with fishers, managers, scientists, conservationists, and other interested groups to build a balanced approach to meet common fisheries goals. This will

ensure best buy-in of any future management measure.

G. Enforce stringent monitoring, control and surveillance mechanisms to restrict unregulated fishing in spawning aggregations that restrain stock recovery.